

Ali Ghebleh

 Ali Ghebleh |  Ali Ghebleh |  ali.ghebleh@fau.de |  +49 178 3348533

Education

Friedrich-Alexander University Erlangen-Nürnberg (FAU) *Oct 2023 – Sep 2026*

MSc Integrated Life Sciences

GPA: 2.3 (Bavarian scale) | Thesis in progress

Supervisors: PD Dr. Claus Metzner, Dr. Patrick Krauss

Sharif University of Technology, Tehran, Iran *Sep 2019 – Feb 2023*

BSc Physics

GPA: 16.59/20 (3.56/4.0); combined GPA including transferred credits: 17.09/20 (3.69/4.0)

Ferdowsi University of Mashhad, Iran *Sep 2018 – Sep 2019*

BSc Physics, first year | GPA: 18.88/20

Admitted to Sharif University on merit; 30 credits transferred

Publications

C. Metzner, **A. Ghebleh**, K. Prebeck, A. Schilling, A. Maier, T. Kinfe, P. Krauss. “Are cortical microcircuits optimized for information flux? A simulation-based reverse engineering study.” *Preprint arXiv:2605.14680*, submitted to *Frontiers in Complex Systems* (2026).

Research Experience

MSc Research — Cognitive Computational Neuroscience Group, FAU *Jan 2025 – May 2026*

Supervisors: PD Dr. Claus Metzner, Dr. Patrick Krauss

Are cortical microcircuits optimized for information flux? *Oct 2025 – May 2026*

- Designed and built the complete simulation pipeline for a Boltzmann machine model of cortical layer 5 microcircuits, including model architecture and hyperparameter selection
- Implemented mutual information and entropy computation analytically (exact, small N) and via Monte Carlo methods (large N)
- Constructed N-Rooks weight matrices as a systematic framework for maximizing information flux
- Ran systematic reverse-engineering experiments across parameter space, identifying two flux-enhancing mechanisms of the embedding network

‘Surviving by Serving’ as a Principle of Selforganization *Jan 2025 – Oct 2025*

- Designed and built a multi-agent simulation of a minimal model economy in which production networks self-organize through purely local feedback, with no centralized control
- Implemented stochastic adaptation mechanisms enabling agents to explore product space and form emergent supply chains

→ Manuscript in preparation

Research Project — Computational Physics Lab, Sharif University of Technology

Mar 2022 – Feb 2023

Supervisor: Dr. Saman Moghimi Araghi

- Extended a biologically-plausible neural network model (Del Papa, Priesemann & Triesch) that achieves criticality via plasticity rules and reinforcement learning; upgraded the network to a more complex architecture

Research Project — Computational Physics Lab, Sharif University of Technology

Sep 2021 – Feb 2022

Supervisors: Dr. Saman Moghimi Araghi, Dr. Sadegh Raeisi

- Applied machine learning methods to detect phase transitions and critical behavior in neuronal avalanche data

Skills

Programming: Python (NumPy, SciPy, Matplotlib), C++ (basic)

Computational Methods: Boltzmann machine simulation, Monte Carlo methods, mutual information and entropy computation, recurrent neural networks, reservoir computing, stochastic systems simulation, neural circuit motif analysis

Machine Learning: Reinforcement learning (epsilon-greedy policy), supervised learning

Languages: Persian (native), English (C1), German (B1)

Awards & Honors

Khwarizmi Youth Award

2016–2017

Nominated for best student research project in Iran; certificate from the Director General of Education, Tehran

International Junior Science Olympiad (IJSO)

2015

International participant, Daegu, South Korea; second place in national team selection competition

References

PD Dr. Claus Metzner

Cognitive Computational Neuroscience Group
FAU Erlangen-Nürnberg
claus.metzner@gmail.com

Dr. Patrick Krauss

Neuromodulation and Neuroprosthetics
University Hospital Mannheim, Heidelberg University
patrick.krauss@fau.de